

Assignment 4

1. There are two barbers in a barber shop they own and operate. They provide two chairs for customers who are waiting to begin a haircut, so the number of customers in the shop varies between 0 and 4. For $n = 0, 1, 2, 3, 4$, the probability P_n that exactly n customers are in the shop is $P_0 = 1/16$, $P_1 = 4/16$, $P_2 = 6/16$, $P_3 = 4/16$, $P_4 = 1/16$

- (a) Calculate L . How would you describe the meaning of L ?
- (b) For each of the possible values of the number of customers in the queueing system, specify how many customers are in the queue. Then calculate L_q . How would you describe the meaning of L_q ?
- (c) Determine the expected number of customers being served.
- (d) Given that an average of 4 customers per hour arrive and stay to receive a haircut, determine W and W_q . Describe these two quantities in meaningful terms.
- (e) Given that two barbers are equally fast in giving haircuts, what is the average duration of a haircut?

2. Suppose that a queueing system has two servers, an exponential interarrival time distribution with a mean of 2 hours, and an exponential service-time distribution with a mean of 2 hours for each server. Furthermore, a customer has just arrived at 12:00 noon.

- (a) What is the probability that the next arrival will come (i) before 1:00 P.M., (ii) between 1:00 and 2:00 P.M., and (iii) after 2:00 P.M.?
- (b) Suppose that no additional customers arrive before 1:00 P.M. Now what is the probability that the next arrival will come between 1:00 and 2:00 P.M.?
- (c) What is the probability that the number of arrivals between 1:00 and 2:00 P.M. will be (i) 0, (ii) 1, and (iii) 2 or more?
- (d) Suppose that both servers are serving customers at 1:00 P.M. What is the probability that neither customer will have service completed (i) before 2:00 P.M., (ii) before 1:10 P.M., and (iii) before 1:01 P.M.?